

Power Query — Pivot & Unpivot Complete Practical Guide

Pivot and Unpivot are **very important Power Query transformations**. The easiest way to understand them is:

Unpivot: Columns → Rows

Pivot: Rows → Columns

1. UNPIVOT

Suppose your Excel data looks like this:

Product	Jan	Feb	Mar
Laptop	25000	28000	30000
Mouse	1200	1500	1700
Keyboard	2200	2500	2600

Here, **Jan, Feb and Mar are columns**.

For Power BI analysis, it is usually better to have:

Product	Month	Sales
Laptop	Jan	25000
Laptop	Feb	28000
Laptop	Mar	30000
Mouse	Jan	1200
Mouse	Feb	1500
Mouse	Mar	1700

This is called **Unpivot**.

2. How to Unpivot

Step 1

Open Power BI.

Home → Get Data → Excel → Transform Data

Step 2

You will see your table in Power Query.

Step 3

Select:

Jan, Feb, Mar

You can select multiple columns using **Ctrl + Click**.

Step 4

Go to:

Transform → Unpivot Columns

You will get:

Product Attribute Value

Laptop	Jan	25000
Laptop	Feb	28000
Laptop	Mar	30000
Mouse	Jan	1200

Product Attribute Value

Mouse	Feb	1500
Mouse	Mar	1700

Step 5

Rename:

Attribute → Month

Value → Sales

Final table:

Product Month Sales

Laptop	Jan	25000
Laptop	Feb	28000
Laptop	Mar	30000
Mouse	Jan	1200
Mouse	Feb	1500
Mouse	Mar	1700

3. Unpivot Other Columns

There is another important option:

Transform → Unpivot Other Columns

Suppose you have:

Product	Category	Jan	Feb	Mar
Laptop	Electronics	25000	28000	30000
Mouse	Accessories	1200	1500	1700

You select:

Product and Category

Then choose:

Unpivot Other Columns

Power Query automatically unpivots Jan, Feb and Mar.

This is **very useful when you have many month columns**.

Exam Tip ☆

If the question says:

Keep Product, Category and Region unchanged and convert all remaining columns into rows.

Use:

Unpivot Other Columns

4. PIVOT

Now suppose you have this data:

Product	Month	Sales
Laptop	Jan	25000
Laptop	Feb	28000
Laptop	Mar	30000

Product Month Sales

Mouse	Jan	1200
Mouse	Feb	1500
Mouse	Mar	1700

You want:

Product	Jan	Feb	Mar
Laptop	25000	28000	30000
Mouse	1200	1500	1700

This is **Pivot**.

5. How to Pivot

Step 1

Select the **Month** column.

Step 2

Go to:

Transform → Pivot Column

Step 3

Power Query opens the Pivot Column window.

You will see:

Column to pivot: Month

Values Column: Sales

Step 4

Choose:

Advanced options → Aggregate Value Function

Select:

Sum

Step 5

Click **OK**.

You get:

Product	Jan	Feb	Mar
Laptop	25000	28000	30000
Mouse	1200	1500	1700

6. Why Does Pivot Ask for Aggregation?

This is extremely important.

Suppose you have:

Product	Month	Sales
Laptop	Jan	25000
Laptop	Jan	30000
Laptop	Feb	28000

There are **two January values** for Laptop.

When Power Query creates:

Product Jan

Laptop ?

Which value should it use?

It needs an aggregation.

You can choose:

- Sum
- Average
- Count
- Min
- Max
- Don't Aggregate

For example:

Sum → 25,000 + 30,000 = 55,000

So:

Product	Jan	Feb
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Laptop	55000	28000
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7. Pivot vs Unpivot

Pivot

Rows → Columns

Creates wider table

Uses a column's values as headers

Often requires aggregation

Transform → Pivot Column

Unpivot

Columns → Rows

Creates longer table

Converts headers into values

Usually doesn't require aggregation

Transform → Unpivot Columns

Remember:

PIVOT = Make columns

UNPIVOT = Make rows

8. Practice Dataset 1 — Basic Unpivot

Copy this table into Excel.

Product	Jan	Feb	Mar	Apr	May	Jun
Laptop	50000	55000	52000	58000	60000	62000
Mouse	5000	5500	6000	5800	6200	6500
Keyboard	8000	8500	9000	9200	9500	9800
Monitor	25000	27000	26000	29000	31000	33000
Printer	15000	16000	15500	17000	18000	19000

Practice Question 1

Using Power Query:

1. Import the table.
2. Unpivot Jan–Jun.
3. Rename `Attribute` to `Month`.
4. Rename `Value` to `Sales`.
5. Change Sales to Decimal/Whole Number.
6. Sort Sales from largest to smallest.

Expected structure:

Product	Month	Sales
Laptop	Jan	50000
Laptop	Feb	55000

Product Month Sales

...

9. Practice Dataset 2 — Unpivot Other Columns

Product	Category	Region	Jan	Feb	Mar	Apr
Laptop	Electronics	East	50000	55000	52000	58000
Mouse	Accessories	West	5000	5500	6000	5800
Keyboard	Accessories	North	8000	8500	9000	9200
Monitor	Electronics	South	25000	27000	26000	29000

Practice Question 2

Using **Unpivot Other Columns**:

1. Keep Product, Category and Region.
2. Convert Jan–Apr into rows.
3. Rename Attribute → Month.
4. Rename Value → Sales.
5. Change Sales to Whole Number.

Important

Don't select Jan–Apr this time.

Select:

Product + Category + Region

Then:

Transform → Unpivot Other Columns

This is an important exam technique.

10. Practice Dataset 3 — Pivot with Duplicate Values

This one is more important for advanced practice.

Product Region Month Sales

Laptop	East	Jan	25000
Laptop	East	Jan	30000
Laptop	East	Feb	28000
Laptop	West	Jan	22000
Laptop	West	Jan	18000
Laptop	West	Feb	25000
Mouse	East	Jan	5000
Mouse	East	Jan	3000
Mouse	East	Feb	6000
Mouse	West	Jan	4500
Mouse	West	Feb	5500

Practice Question 3

Create a Pivot table where:

Rows:

- Product
- Region

Columns:

- Month

Values:

- Sales

Use:

Aggregation = Sum

Expected result:

Product	Region	Jan	Feb
Laptop	East	55000	28000
Laptop	West	40000	25000
Mouse	East	8000	6000
Mouse	West	4500	5500

11. Practice Dataset 4 — Advanced Pivot

Department	Employee	Quarter	Sales
IT	Amit	Q1	100000
IT	Amit	Q2	120000
IT	Amit	Q3	110000
IT	Amit	Q4	140000
HR	Riya	Q1	60000
HR	Riya	Q2	70000
HR	Riya	Q3	75000
HR	Riya	Q4	80000
Sales	Rahul	Q1	150000
Sales	Rahul	Q2	170000

Department Employee Quarter Sales

Sales	Rahul	Q3	160000
Sales	Rahul	Q4	190000

Practice Question 4

Create:

Department Employee Q1 Q2 Q3 Q4

Use:

Pivot Column = Quarter

Values Column = Sales

Aggregation = Sum

12. Advanced Practice — Average

Use the Dataset 3.

Create a Pivot where:

Rows: Product + Region

Columns: Month

Values: Sales

But this time use:

Aggregation = Average

Don't use Sum.

This will teach you why aggregation matters.

13. Advanced Practice — Count

Create a new table:

Department Employee Status

IT	Amit	Present
IT	Amit	Absent
IT	Rahul	Present
HR	Riya	Present
HR	Riya	Present
Sales	Rahul	Absent

Question

Pivot the **Status** column.

Use **Count** as the aggregation.

Your goal is to create something like:

Department Employee Present Absent

IT	Amit	1	1
IT	Rahul	1	0
HR	Riya	2	0
Sales	Rahul	0	1

14. Very Important Real-World Example

Suppose an Excel file comes like this:

Customer	Jan Sales	Feb Sales	Mar Sales	Apr Sales
A	10000	12000	15000	14000
B	8000	9000	10000	11000

For Power BI, you usually want:

Customer	Month	Sales
A	Jan Sales	10000
A	Feb Sales	12000
A	Mar Sales	15000
A	Apr Sales	14000
B	Jan Sales	8000
B	Feb Sales	9000
B	Mar Sales	10000
B	Apr Sales	11000

So you would use **Unpivot**.

This structure is much easier for:

- Line charts
- Month-wise analysis
- Slicers
- Date analysis
- Grouping
- Measures
- Time-series dashboards

15. Exam Questions You Should Practice

Basic

Q1. Unpivot monthly sales columns into Month and Sales.

Q2. Pivot Month into separate columns.

Q3. Unpivot selected columns.

Q4. Use Unpivot Other Columns.

Intermediate

Q5. Pivot Month using Sum.

Q6. Pivot Month using Average.

Q7. Pivot Status using Count.

Q8. Pivot with duplicate records.

Advanced

Q9. Pivot using multiple identifying columns such as Product + Region.

Q10. Unpivot a table containing Product, Category, Region and 12 monthly columns.

Q11. Handle duplicate Product-Month combinations using Sum.

Q12. Convert quarterly data Q1–Q4 into columns using Pivot.

16. What You Should Remember for the Exam ☆

When you see:

"Convert columns into rows."

Think:

UNPIVOT

When you see:

"Convert rows into columns."

Think:

PIVOT

When you see:

"Keep these columns and convert all remaining columns into rows."

Think:

UNPIVOT OTHER COLUMNS

When you see:

"There are duplicate values and you need total."

Think:

PIVOT → Sum

When you see:

"There are duplicate values and you need average."

Think:

PIVOT → Average

When you see:

"Count the occurrences."

Think:

PIVOT → Count

Power Query: Fill Up, Fill Down, Pivot & Unpivot

These four operations are important for Power BI practical exams. We'll learn them together using practical tables.

1. Fill Down

Fill Down copies the value from the cell **above** into the blank cells below it.

Example

Suppose your data is:

Department Employee Salary

IT	Amit	40000
null	Rahul	45000
null	Priya	42000
HR	Riya	38000
null	Suman	40000

Here, the blank Department cells should belong to the department above them.

How to do it

1. Open the table in **Power Query**.
2. Select the **Department** column.
3. Go to:

Transform → Fill → Down

Result

Department Employee Salary

IT	Amit	40000
IT	Rahul	45000

Department Employee Salary

IT	Priya	42000
HR	Riya	38000
HR	Suman	40000

Remember

Fill Down = Take value from **ABOVE** → move **DOWN**

2. Fill Up

Fill Up does the opposite.

It copies the value from the **cell below** into blank cells above it.

Example

Department Employee

null	Amit
null	Rahul
IT	Priya
null	Riya
HR	Suman

Select **Department**.

Go to:

Transform → **Fill** → **Up**

Result:

Department Employee

IT	Amit
IT	Rahul
IT	Priya
HR	Riya
HR	Suman

Remember

Fill Up = Take value from BELOW → move UP

3. Fill Down vs Fill Up

Fill Down

Uses value above

Fills blanks downward

Transform → Fill → Down

Fill Up

Uses value below

Fills blanks upward

Transform → Fill → Up

Easy memory trick

DOWN ↓ = Look UP ↑

UP ↑ = Look DOWN ↓

4. Pivot

Now let's connect this with Pivot.

Suppose we have:

Product Month Sales

Laptop	Jan	50000
Laptop	Feb	55000
Laptop	Mar	60000
Mouse	Jan	5000
Mouse	Feb	6000
Mouse	Mar	7000

We want:

Product	Jan	Feb	Mar
Laptop	50000	55000	60000
Mouse	5000	6000	7000

Steps

1. Select **Month**.
2. Go to **Transform** → **Pivot Column**.
3. Values Column = **Sales**.
4. Aggregation = **Sum**.
5. Click **OK**.

5. Unpivot

Unpivot does the reverse.

Starting with:

Product	Jan	Feb	Mar
Laptop	50000	55000	60000

Product	Jan	Feb	Mar
Mouse	5000	6000	7000

Select:

Jan, Feb, Mar

Then:

Transform → Unpivot Columns

Result:

Product	Attribute	Value
Laptop	Jan	50000
Laptop	Feb	55000
Laptop	Mar	60000
Mouse	Jan	5000
Mouse	Feb	6000
Mouse	Mar	7000

Rename:

Attribute → Month

Value → Sales

6. Fill + Pivot Practical Example

Now let's combine the concepts.

Copy this table into Excel:

Department Employee Month Sales

IT	Amit	Jan	50000
null	Amit	Feb	55000
null	Amit	Mar	60000
HR	Riya	Jan	40000
null	Riya	Feb	45000
null	Riya	Mar	48000
Sales	Rahul	Jan	70000
null	Rahul	Feb	75000
null	Rahul	Mar	80000

Practice Question 1

Clean and transform this table:

Task 1

Fill the missing Department values using **Fill Down**.

You should get:

Department Employee Month Sales

IT	Amit	Jan	50000
IT	Amit	Feb	55000
IT	Amit	Mar	60000
HR	Riya	Jan	40000
HR	Riya	Feb	45000
HR	Riya	Mar	48000

Department Employee Month Sales

Sales	Rahul	Jan	70000
Sales	Rahul	Feb	75000
Sales	Rahul	Mar	80000

Task 2

Now **Pivot Month**.

Use:

Pivot Column: Month

Values Column: Sales

Aggregation: Sum

Expected result:

Department	Employee	Jan	Feb	Mar
IT	Amit	50000	55000	60000
HR	Riya	40000	45000	48000
Sales	Rahul	70000	75000	80000

7. Fill Up + Unpivot Practice

Now practice **Fill Up**.

Copy this:

Category	Product	Jan	Feb	Mar
null	Laptop	50000	55000	60000

Category	Product	Jan	Feb	Mar
null	Mouse	5000	6000	7000
Electronics	Keyboard	8000	9000	10000
null	Monitor	25000	27000	30000
Accessories	Printer	15000	16000	18000
null	Scanner	10000	12000	14000

Practice Question 2

Task 1: Use **Fill Up** on Category.

Expected:

Category	Product
Electronics	Laptop
Electronics	Mouse
Electronics	Keyboard
Accessories	Monitor
Accessories	Printer
Accessories	Scanner

Task 2: Unpivot Jan, Feb and Mar.

Expected structure:

Category	Product	Month	Sales
Electronics	Laptop	Jan	50000
Electronics	Laptop	Feb	55000
Electronics	Laptop	Mar	60000

Category Product Month Sales

...

8. Advanced Practice: Fill Down + Unpivot + Pivot

Use this table:

Region	Category	Product	Jan	Feb	Mar	Apr
East	null	Laptop	50000	55000	60000	65000
null	Electronics	Mouse	5000	6000	7000	7500
null	null	Keyboard	8000	9000	10000	11000
West	Accessories	Printer	15000	16000	18000	20000
null	null	Scanner	10000	12000	14000	15000

Practice Question 3

Perform these operations **in order**:

Step 1

Fill Down **Region**.

Step 2

Fill Down **Category**.

Step 3

Unpivot:

Jan, Feb, Mar, Apr

Step 4

Rename:

Attribute → **Month**

Value → **Sales**

Step 5

Change Sales to **Whole Number**.

Step 6

Create a Pivot:

Rows: Region, Category, Product

Columns: Month

Values: Sales

Aggregation: Sum

This single exercise gives you practice with:

Fill Down → **Unpivot** → **Rename** → **Data Type** → **Pivot**

9. Important Exam Situations

Situation 1

"Department is entered only once and blank cells below should contain the same department."

Use:

Fill Down

Situation 2

"Category is present below the blank cells. Fill the blank cells using the value below."

Use:

Fill Up

Situation 3

"Convert Jan, Feb, Mar and Apr columns into rows."

Use:

Unpivot

Situation 4

"Convert Month values into separate columns."

Use:

Pivot

Situation 5

"There are multiple records for the same Product and Month. Calculate total sales."

Use:

Pivot → Aggregation = Sum

10. One Important Rule

The order of transformations can matter.

For example, if your source looks like:

Department → blank → blank → next Department

first use:

Fill Down

and then perform:

Unpivot

Don't blindly unpivot first. Make sure your identifying/category columns are properly filled before restructuring the data.

Final Memory Table

Requirement	Power Query Operation
Copy value above into blanks	Fill Down
Copy value below into blanks	Fill Up
Columns → Rows	Unpivot
Rows → Columns	Pivot
Duplicate values during Pivot	Aggregation
Total during Pivot	Sum
Average during Pivot	Average
Number of records	Count

Best practice for your exam: Do Practice Question 3 completely without looking at the steps. It combines the four transformations and is close to the type of Power Query practical task you should be able to solve independently.